CHAPTER 2 · CORE THEORY

Flow Kinematics

Fluid Mechanics — 2nd Year Bachelor's Degree in Aerospace Engineering (UC3M)

Reference sources: Notes.pdf (Chapter 2, pp. 9–24) | slides_Chapters1-2.pdf (2.0–2.24) · Profs. Antonio L. Sánchez & Javier Rodríguez-Rodríguez

1. Vector and Tensor Formulas in Orthogonal Curvilinear Coordinates

The rigorous analysis of aerospace flows requires working in orthogonal coordinate systems adapted to the geometry of the flow. Consider a local orthonormal unit-vector basis $(\vec{e}_1, \vec{e}_2, \vec{e}_3)$ at each point in space.

Scale Factors (h_1, h_2, h_3) and Differential Line Element

The differential displacement vector is universally expressed as:

$$d\vec{x} = h_1 dx_1 \vec{e}_1 + h_2 dx_2 \vec{e}_2 + h_3 dx_3 \vec{e}_3$$

Coordinate System	**Coordinates** (x_1, x_2, x_3)	**Scale Factors** (h_1, h_2, h_3)	**Differential Line Element** $d\vec{x}$
Cartesian	(x, y, z)	$h_1 = 1,\ h_2 = 1,\ h_3 = 1$	(dx, dy, dz)
Cylindrical	(r, θ, z)	$h_1 = 1,\ h_2 = r,\ h_3 = 1$	$(dr, rd\theta, dz)$
Spherical	(r, θ, ϕ)	$h_1 = 1,\ h_2 = r,\ h_3 = r\sin\theta$	$(dr, rd\theta, r\sin\theta d\phi)$

Fundamental Definitions of Tensor Algebra

- **Second-order tensor ($\bar{\bar{A}}$):** Linear operator transforming a vector $\vec{a}$ into another vector $\vec{b} = \bar{\bar{A}} \cdot \vec{a}$, with components $b_i = \sum_j A_{ij} a_j$.
- **Transpose tensor ($\bar{\bar{A}}^T$):** Satisfies $(\bar{\bar{A}}^T)_{ij} = A_{ji}$ and the identity $\vec{a} \cdot \bar{\bar{A}} = \bar{\bar{A}}^T \cdot \vec{a}$.
- **Identity tensor ($\bar{\bar{I}}$):** $I_{ii} = 1$, $I_{ij} = 0$ $(i \neq j)$, such that $\bar{\bar{I}} \cdot \vec{a} = \vec{a}$.
- **Dyadic product of two vectors ($\vec{a}\vec{b}$):** Yields a tensor with components $(\vec{a}\vec{b})_{ij} = a_i b_j$.
- **Double contraction of tensors ($\bar{\bar{A}} : \bar{\bar{B}}$):** Produces a scalar defined by $\bar{\bar{A}} : \bar{\bar{B}} = \sum_i \sum_j A_{ij} B_{ij}$. (Distinct from the tensor dot product $\bar{\bar{A}} \cdot \bar{\bar{B}}$, whose components are $(\bar{\bar{A}} \cdot \bar{\bar{B}})_{ij} = \sum_k A_{ik} B_{kj}$).
- **Cross product of a vector and a tensor ($\vec{a} \wedge \bar{\bar{A}}$):**

$$\vec{a} \wedge \bar{\bar{A}} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \bar{\bar{A}} \qquad [\text{Eq. } 2.2]$$

Invariant Differential Operators (Notes.pdf, Eqs. 2.3–2.8)

Defining the metric determinant $h = h_1 h_2 h_3$, the differential operators adopt their general coordinate-invariant formulation:

1. Gradient of a Scalar Φ:

$$\nabla \Phi = \left(\frac{1}{h_1} \frac{\partial \Phi}{\partial x_1}, \frac{1}{h_2} \frac{\partial \Phi}{\partial x_2}, \frac{1}{h_3} \frac{\partial \Phi}{\partial x_3} \right) \qquad [\text{Eq. } 2.3]$$

2. Laplacian of a Scalar Φ:

$$\nabla^2 \Phi = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial x_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial \Phi}{\partial x_1} \right) + \frac{\partial}{\partial x_2} \left(\frac{h_1 h_3}{h_2} \frac{\partial \Phi}{\partial x_2} \right) + \frac{\partial}{\partial x_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial \Phi}{\partial x_3} \right) \right] \qquad [\text{Eq. } 2.4]$$

3. Divergence of a Vector $\vec{a}$:

$$\nabla \cdot \vec{a} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial x_1} (h_2 h_3 a_1) + \frac{\partial}{\partial x_2} (h_1 h_3 a_2) + \frac{\partial}{\partial x_3} (h_1 h_2 a_3) \right] \qquad [\text{Eq. } 2.5]$$

4. Curl of a Vector $\vec{a}$:

$$\nabla \wedge \vec{a} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1\vec{e}_1 & h_2\vec{e}_2 & h_3\vec{e}_3 \\ \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_3} \\ h_1 a_1 & h_2 a_2 & h_3 a_3 \end{vmatrix} \qquad [\text{Eq. } 2.6]$$

5. Gradient of a Vector Field ($\nabla \vec{a}$):

$$(\nabla \vec{a})_{ii} = \frac{1}{h_i}\frac{\partial a_i}{\partial x_i} + \sum_{k \neq i} \frac{a_k}{h_i h_k}\frac{\partial h_i}{\partial x_k}, \qquad (\nabla \vec{a})_{ij} = \frac{1}{h_i}\frac{\partial a_j}{\partial x_i} - \frac{a_i}{h_i h_j}\frac{\partial h_i}{\partial x_j} \ (i \neq j) \qquad [\text{Eq. } 2.7]$$

6. Divergence of a Tensor ($\nabla \cdot \bar{\bar{A}}$):

$$(\nabla \cdot \bar{\bar{A}})_i = \frac{h_i}{h} \sum_j \frac{\partial}{\partial x_j}\left(\frac{h A_{ij}}{h_i h_j}\right) + \sum_j \frac{A_{ij} + A_{ji}}{h_i h_j}\frac{\partial h_i}{\partial x_j} - \sum_j \frac{A_{jj}}{h_i h_j}\frac{\partial h_j}{\partial x_i} \qquad [\text{Eq. } 2.8]$$

General Gauss Divergence Theorem (Notes.pdf, Eq. 2.9)

$$\int_\Sigma (\vec{n} \circ \phi)\, d\sigma = \int_V (\nabla \circ \phi)\, dV$$

where ϕ represents a scalar, vector, or tensor field, V is the domain bounded by the closed surface Σ, and the symbol $\circ$ denotes the appropriate scalar, dyadic, or tensor product.

2. Kinematic Descriptions: Eulerian vs. Lagrangian

There are two conceptual frameworks to describe the motion of a continuum fluid:

- **Lagrangian Description:** Adopts a discrete perspective, tracking individual fluid particles identified by their initial position $\vec{x}_0$ at time t_0. The flow is described via trajectories:

$$\vec{x} = \vec{x}_T(\vec{x}_0, t) \qquad [\text{Eq. } 2.10]$$

  Velocity and acceleration are obtained by direct time differentiation: $\vec{v} = \frac{\partial \vec{x}_T}{\partial t}$, $\vec{a} = \frac{\partial^2 \vec{x}_T}{\partial t^2}$.
- **Eulerian Description:** Treats velocity as a continuous vector field dependent on the fixed spatial position $\vec{x}$ and time t:

$$\vec{v} = \vec{v}(\vec{x}, t)$$

  This is the framework systematically adopted in fluid mechanics, as it substantially simplifies the formulation of integral and differential conservation laws.

Preliminary Kinematic Concepts

- **Uniform Flow:** Spatial variations of all properties are identically zero. The velocity field depends solely on time: $\vec{v} = \vec{v}(t)$.
- **Steady Flow:** Flow properties at every spatial point do not vary with time: $\frac{\partial \vec{v}}{\partial t} = 0 \implies \vec{v} = \vec{v}(\vec{x})$.
- **Frame Relativity:** Whether a flow is steady or unsteady depends on the chosen frame of reference. Around an aircraft cruising at constant velocity, the flow is unsteady to a ground-fixed observer ($\vec{v}(\vec{x}, t)$), but strictly steady to an observer attached to the aircraft or in a wind tunnel.
- **Stagnation Point:** A spatial point where fluid velocity vanishes simultaneously in all three components:

$$\vec{v}(\vec{x}, t) = 0 \qquad [\text{Eq. } 2.11]$$

  The spatial location of stagnation points changes under a change of reference frame.

3. Trajectories and Pathlines

A fluid particle initially located at $\vec{x}_0$ at time t_0 evolves in space following the trajectory dictated by the Eulerian velocity field:

$$\frac{d\vec{x}}{dt} = \vec{v}(\vec{x}, t), \qquad \text{with initial condition } \vec{x}(t_0) = \vec{x}_0 \qquad [\text{Eq. 2.12}]$$

In general curvilinear coordinates, this system of ordinary differential equations reads:

$$dt = \frac{h_1 dx_1}{v_1(x_1, x_2, x_3, t)} = \frac{h_2 dx_2}{v_2(x_1, x_2, x_3, t)} = \frac{h_3 dx_3}{v_3(x_1, x_2, x_3, t)} \qquad [\text{Eq. 2.13}]$$

The time-dependent solution represents the trajectory $\vec{x} = \vec{x}_T(\vec{x}_0, t)$ (Eq. 2.14 in Cartesian coordinates). The trajectory contains complete information regarding both the geometric spatial curve and the instantaneous speed at which the particle traverses it.

Eliminating the time parameter t from the parametric trajectory equations yields the geometric **pathlines**, expressed as the intersection of two implicit surfaces:

$$f(x_0, y_0, z_0, x, y, z) = 0 \quad \text{and} \quad g(x_0, y_0, z_0, x, y, z) = 0 \qquad [\text{Eq. 2.15}]$$

4. Fluid Lines, Fluid Surfaces, and Fluid Volumes

Fluid Line: Fluid particles that lie along a parameterized curve $\vec{x} = \vec{x}_l(\lambda)$ at the initial instant t_0 will continue to form a continuous curve at any subsequent instant:

$$\vec{x} = \vec{x}_T(\vec{x}_l(\lambda), t) \qquad [\text{Eq. 2.17}]$$

Fluid Surface: Similarly, particles forming an initial surface $\vec{x} = \vec{x}_s(\alpha, \beta)$ evolve into a fluid surface $\vec{x} = \vec{x}_T(\vec{x}_s(\alpha, \beta), t)$ (Eq. 2.19), expressible implicitly as $f(\vec{x}, t) = 0$ (Eq. 2.20).

Kinematic Boundary Condition for a Fluid Surface: Because particles belonging to the surface never leave it, the value $f(\vec{x}, t) = 0$ is conserved along the motion of the particles, requiring:

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \vec{v} \cdot \nabla f = 0$$

Fluid Volume: The finite volume of fluid bounded by a closed fluid surface. Because fluid velocity is defined as the center-of-mass velocity of the molecular cluster, **mass cannot cross a fluid surface**. Consequently, the mass of a fluid volume remains strictly constant in time:

$$\frac{d}{dt}\int_{V_f(t)} \rho \, dV = 0 \qquad \text{[First Principle of Conservation]}$$

5. Streamlines, Stream Surfaces, and Streamtubes

The **streamlines** at a given time t are curves whose tangent at every point is collinear with the instantaneous velocity vector:

$$d\vec{x} \wedge \vec{v} = 0 \iff \frac{h_1 dx_1}{v_1(x_1, x_2, x_3, t)} = \frac{h_2 dx_2}{v_2(x_1, x_2, x_3, t)} = \frac{h_3 dx_3}{v_3(x_1, x_2, x_3, t)} \qquad \text{[Eq. 2.21]}$$

- In Cartesian coordinates: $\frac{dx}{v_x} = \frac{dy}{v_y} = \frac{dz}{v_z}$.
- In Cylindrical coordinates: $\frac{dr}{v_r} = \frac{r d\theta}{v_\theta} = \frac{dz}{v_z}$.
- In Spherical coordinates: $\frac{dr}{v_r} = \frac{r d\theta}{v_\theta} = \frac{r \sin\theta d\phi}{v_\phi}$.

Fundamental Physical Distinction: Trajectory vs. Streamline

Although equations (2.13) and (2.21) appear algebraically similar, they represent completely distinct physical and mathematical concepts:

- In a **trajectory**, time t is the dynamic integration variable of an initial value (Cauchy) problem tracking an individual particle through history.
- In a **streamline**, time t is "frozen"; it acts purely as a fixed parameter defining an instantaneous snapshot of the vector field in space.
- **Coincidence Theorem:** Trajectories, streamlines, and pathlines **coincide** in steady flows ($\vec{v} = \vec{v}(\vec{x})$), or more generally, in flows where the direction of the velocity vector is time-invariant: $\vec{v}(\vec{x}, t) = f(t)\vec{V}(\vec{x})$.

Stream Surface: The surface formed by all streamlines passing through a given directrix curve. Because $\vec{v}$ is everywhere tangent to the surface, no convective flux crosses a stream surface ($\vec{v} \cdot \vec{n} = 0$).

Streamtube: A stream surface generated by a closed directrix curve. It acts as an "impermeable fluid conduit" through which mass flows confined.

6. Material Derivative and Acceleration Field

To quantify the rate of change of an intensive scalar property $\phi(\vec{x}, t)$ (density, pressure, temperature) experienced by a fluid particle during its differential displacement $d\vec{x} = \vec{v}dt$:

$$d\phi = \phi(\vec{x} + d\vec{x}, t + dt) - \phi(\vec{x}, t) = \frac{\partial \phi}{\partial t}dt + d\vec{x} \cdot \nabla \phi \qquad [\text{Eq. } 2.22]$$

Dividing by dt yields the **material derivative operator**:

$$\boxed{\frac{\mathbf{D}\phi}{\mathbf{D}\mathbf{t}} = \underbrace{\frac{\boldsymbol{\partial}\phi}{\boldsymbol{\partial}\mathbf{t}}}_{\text{Local Rate of Change}} + \underbrace{\vec{\mathbf{v}} \cdot \boldsymbol{\nabla}\phi}_{\text{Convective Rate of Change}}} \qquad [\text{Eqs. } 2.23, 2.24]$$

The Acceleration Vector of a Fluid Particle

Applying the material operator to the velocity field $\vec{v}(\vec{x}, t)$:

$$\vec{a} = \frac{D\vec{v}}{Dt} = \frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot (\nabla \vec{v}) \qquad [\text{Eq. } 2.25]$$

Universal Intrinsic Expression (Notes.pdf, Eq. 2.26):

Via vector calculus identities, acceleration can be expressed in coordinate-free form as:

$$\boxed{\vec{\mathbf{a}} = \frac{\boldsymbol{\partial}\vec{\mathbf{v}}}{\boldsymbol{\partial}\mathbf{t}} + \boldsymbol{\nabla}\left(\frac{|\vec{\mathbf{v}}|^{\mathbf{2}}}{\mathbf{2}}\right) - \vec{\mathbf{v}} \wedge (\boldsymbol{\nabla} \wedge \vec{\mathbf{v}})}$$

where $-\vec{v} \wedge (\nabla \wedge \vec{v}) = -\vec{v} \wedge \vec{\omega}$ is the Lamb acceleration due to vorticity.

In Cartesian coordinates, each acceleration component equals the material derivative of the corresponding velocity component:

$$a_i = \frac{Dv_i}{Dt} = \frac{\partial v_i}{\partial t} + \sum_j v_j \frac{\partial v_i}{\partial x_j} = \frac{\partial v_i}{\partial t} + v_x \frac{\partial v_i}{\partial x} + v_y \frac{\partial v_i}{\partial y} + v_z \frac{\partial v_i}{\partial z} \qquad [\text{Eq. } 2.27]$$

Analytical note: This direct component-by-component decomposition does not hold in curvilinear coordinates due to the spatial derivatives of the local basis unit vectors.

Acceleration in Non-Inertial Reference Frames

If the reference frame undergoes translational acceleration $\vec{a}_0(t)$ and rotates with angular velocity $\vec{\Omega}(t)$, the apparent acceleration of the system is:

$$\vec{a}_s = \vec{a}_0 + \frac{d\vec{\Omega}}{dt} \wedge \vec{x} + \vec{\Omega} \wedge (\vec{\Omega} \wedge \vec{x}) + 2\vec{\Omega} \wedge \vec{v} \qquad [\text{Eq. } 2.28]$$

7. Circulation, Vorticity, and Irrotational Flow

The **circulation** Γ along an oriented curve L is defined as the line integral:

$$\Gamma = \int_L \vec{v} \cdot d\vec{l} = \int_{\lambda_1}^{\lambda_2} \vec{v}(\vec{x}_l(\lambda), t) \cdot \frac{d\vec{x}_l}{d\lambda} d\lambda \qquad [\text{Eqs. } 2.29, 2.31]$$

For a closed curve L, **Stokes' Theorem** states:

$$\mathbf{\Gamma} = \oint_{\mathbf{L}} \vec{\mathbf{v}} \cdot \mathbf{d}\vec{\mathbf{l}} = \int_{\mathbf{\Sigma}} (\boldsymbol{\nabla} \wedge \vec{\mathbf{v}}) \cdot \vec{\mathbf{n}} \, \mathbf{d}\sigma \qquad [\text{Eq. } 2.32]$$

where Σ is any open surface bounded by L, and the **vorticity vector** is defined as:

$$\vec{\omega} = \boldsymbol{\nabla} \wedge \vec{\mathbf{v}}$$

For a planar differential element of area $d\sigma$ normal to $\vec{n}$, $\oint_L \vec{v} \cdot d\vec{l} = (\nabla \wedge \vec{v}) \cdot \vec{n} d\sigma$ (Eq. 2.33). Hence, $(\nabla \wedge \vec{v}) \cdot \vec{n}$ physically represents the **circulation per unit area oriented along** $\vec{n}$.

Irrotational Flow and Velocity Potential Function (ϕ)

A flow is irrotational when $\vec{\omega} = \nabla \wedge \vec{v} \equiv 0$. Under such conditions, there exists a scalar potential function $\phi(\vec{x}, t)$ such that:

$$\vec{\mathbf{v}} = \boldsymbol{\nabla}\phi \qquad [\text{Eq. } 2.34]$$

> **Domain Topology and Lift Generation (Notes.pdf, p. 16)**
>
> If circulation along every closed curve is zero, the flow is necessarily irrotational. However, **the converse is not universally true**: in multiply-connected domains (such as flow around a 2D airfoil), closed curves exist that cannot be contracted to a point within the fluid domain. Along such curves, circulation $\Gamma \neq 0$ even though $\nabla \wedge \vec{v} = 0$ everywhere throughout the exterior fluid, generating the aerodynamic lift force on airfoils (Kutta-Joukowski Theorem: $L' = \rho_\infty V_\infty \Gamma$).

8. Convective Flux of Extensive Properties

Considering a fluid property $\phi(\vec{x}, t)$ expressed per unit volume, the net transport rate crossing a fixed surface Σ_0 per unit time is the **convective flux**:

$$\boldsymbol{\Phi}_{\text{conv}} = \int_{\boldsymbol{\Sigma_0}} \phi\, \vec{\mathbf{v}} \cdot \vec{\mathbf{n}}\, \mathbf{d}\sigma \qquad [\text{Eq. } 2.35]$$

Transported Quantity	**Volumetric Density (ϕ)**	**Specific Convective Flux ($\phi\vec{v}$)**	**Total Flux across Σ_0**
Volume	$\phi = 1$	$\vec{v}$ (volume flux vector)	$Q = \int_{\Sigma_0} \vec{v} \cdot \vec{n}\, d\sigma$ (Volumetric flow rate, Eq. 2.36)
Mass	$\phi = \rho$	$\rho\vec{v}$ (mass flux vector)	$\dot{m} = \int_{\Sigma_0} \rho\vec{v} \cdot \vec{n}\, d\sigma$ (Mass flow rate)
Linear Momentum	$\phi = \rho\vec{v}$	$\rho\vec{v}\vec{v}$ (momentum flux tensor)	$\int_{\Sigma_0} \rho\vec{v}(\vec{v} \cdot \vec{n})\, d\sigma$
Total Energy	$\phi = \rho\left(e + \frac{\lvert\vec{v}\rvert^2}{2}\right)$	$\rho\left(e + \frac{\lvert\vec{v}\rvert^2}{2}\right)\vec{v}$	$\dot{E} = \int_{\Sigma_0} \rho\left(e + \frac{\lvert\vec{v}\rvert^2}{2}\right)\vec{v} \cdot \vec{n}\, d\sigma$

For a closed surface enclosing volume V_0, the Gauss Divergence Theorem states:

$$\int_{\Sigma_0} \phi\, \vec{v} \cdot \vec{n}\, d\sigma = \int_{V_0} \nabla \cdot (\phi\vec{v})\, dV \qquad [\text{Eq. } 2.37]$$

Therefore, $\nabla \cdot \vec{v}$ represents the rate of volume loss per unit volume (**volumetric expansion or dilation rate**). For a perfect liquid or incompressible fluid whose density is strictly constant:

$$\boldsymbol{\nabla \cdot \vec{\mathbf{v}} = 0} \qquad [\text{Eq. } 2.38]$$

For a moving control surface $\Sigma_c(t)$ with velocity $\vec{v}_c$, the flux is evaluated using the relative velocity: $\int_{\Sigma_c} \phi(\vec{v} - \vec{v}_c) \cdot \vec{n}\, d\sigma$ (Eq. 2.39).

9. The Stream Function (ψ) for Planar Flows

For an incompressible two-dimensional planar flow in the Cartesian $x - y$ plane, the zero-divergence condition takes the form:

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0 \qquad [\text{Eq. } 2.40]$$

The **stream function** $\psi(x, y, t)$ is defined by:

$$\mathbf{v_x} = \frac{\boldsymbol{\partial \psi}}{\boldsymbol{\partial y}}, \qquad \mathbf{v_y} = -\frac{\boldsymbol{\partial \psi}}{\boldsymbol{\partial x}} \qquad [\text{Eq. } 2.41]$$

Mathematical and Physical Properties of the Stream Function

1. Level curves $\psi = \text{constant}$ are Streamlines:

Differentiating ψ along a level curve:

$$d\psi = \frac{\partial \psi}{\partial x}dx + \frac{\partial \psi}{\partial y}dy = -v_y dx + v_x dy = 0 \implies \frac{\mathbf{dx}}{\mathbf{v_x}} = \frac{\mathbf{dy}}{\mathbf{v_y}} \qquad [\text{Eqs. } 2.42, 2.43]$$

which is identical to the differential equation of streamlines.

2. The difference $\Delta\psi = \psi_2 - \psi_1$ measures the Volume Flow Rate:

Integrating $d\psi$ between two streamlines 1 and 2 along an arbitrary curve with unit normal $\vec{n}$:

$$\psi_2 - \psi_1 = \int_1^2 d\psi = \int_1^2 (-v_y dx + v_x dy) = \int_{\mathbf{1}}^{\mathbf{2}} \vec{\mathbf{v}} \cdot \vec{\mathbf{n}}\, \mathbf{dl} = \mathbf{Q'} \qquad [\text{Eqs. } 2.44, 2.45]$$

where Q' is the volumetric flow rate per unit depth normal to the flow plane.

10. Kinematics of Deformation: Strain Rate and Rotation Tensors

The force that adjacent fluid elements exert on one another is proportional to the rate at which the fluid deforms. To first order, the velocity difference across a differential fluid element $d\vec{x}$ is given by:

$$d\vec{v} = d\vec{x} \cdot \nabla\vec{v} \qquad \text{[Eq. 2.46]}$$

Helmholtz Decomposition (Notes.pdf, Eqs. 2.47–2.52)

The velocity gradient tensor $\nabla\vec{v}$ is decomposed uniquely into a symmetric part $\bar{\bar{T}}_d$ and an antisymmetric part $\bar{\bar{T}}_r$:

$$\boldsymbol{\nabla\vec{\mathbf{v}}} = \underbrace{\mathbf{\frac{1}{2}(\nabla\vec{\mathbf{v}} + \nabla\vec{\mathbf{v}}^{T})}}_{\bar{\bar{\mathbf{T}}}_{\mathbf{d}}\ \text{(Rate-of-strain tensor)}} + \underbrace{\mathbf{\frac{1}{2}(\nabla\vec{\mathbf{v}} - \nabla\vec{\mathbf{v}}^{T})}}_{\bar{\bar{\mathbf{T}}}_{\mathbf{r}}\ \text{(Rate-of-rotation tensor)}} \qquad \text{[Eq. 2.47]}$$

A. Rate-of-Rotation Tensor ($\bar{\bar{T}}_r$) and Rigid-Body Rotation:

$$\bar{\bar{T}}_r = \frac{1}{2}\begin{bmatrix} 0 & \omega_3 & -\omega_2 \\ -\omega_3 & 0 & \omega_1 \\ \omega_2 & -\omega_1 & 0 \end{bmatrix} \qquad \text{[Eq. 2.50]}$$

The action of $\bar{\bar{T}}_r$ on the element $d\vec{x}$ produces rigid-body rotation:

$$d\vec{v}_r = d\vec{x} \cdot \bar{\bar{T}}_r = \frac{1}{2}(\nabla \wedge \vec{v}) \wedge d\vec{x} = \mathbf{\frac{1}{2}\vec{\omega} \wedge d\vec{x}} \qquad \text{[Eq. 2.52]}$$

with local angular velocity $\vec{\Omega}_f = \frac{1}{2}\vec{\omega}$. If $\bar{\bar{T}}_d \equiv 0$, the fluid moves purely as a rigid body (pure translation + rigid rotation).

B. Rate-of-Strain Tensor ($\bar{\bar{T}}_d$):

$$\bar{\bar{T}}_d = \begin{bmatrix} \frac{\partial v_1}{\partial x_1} & \frac{1}{2}\left(\frac{\partial v_2}{\partial x_1} + \frac{\partial v_1}{\partial x_2}\right) & \frac{1}{2}\left(\frac{\partial v_3}{\partial x_1} + \frac{\partial v_1}{\partial x_3}\right) \\ \frac{1}{2}\left(\frac{\partial v_2}{\partial x_1} + \frac{\partial v_1}{\partial x_2}\right) & \frac{\partial v_2}{\partial x_2} & \frac{1}{2}\left(\frac{\partial v_3}{\partial x_2} + \frac{\partial v_2}{\partial x_3}\right) \\ \frac{1}{2}\left(\frac{\partial v_3}{\partial x_1} + \frac{\partial v_1}{\partial x_3}\right) & \frac{1}{2}\left(\frac{\partial v_3}{\partial x_2} + \frac{\partial v_2}{\partial x_3}\right) & \frac{\partial v_3}{\partial x_3} \end{bmatrix} \qquad \text{[Eq. 2.48]}$$

Writing $d\vec{x} = \vec{n}ds$, the strain rate velocity increment is $d\vec{v}_d = \bar{\bar{T}}_d \cdot \vec{n}ds$ (Eq. 2.53):

- **Unit rate of extension in direction $\vec{n}$:** $\dot{\epsilon}_n = \vec{n} \cdot \bar{\bar{T}}_d \cdot \vec{n}$.
- **Shear strain rate:** $[\bar{\bar{T}}_d \cdot \vec{n} - (\vec{n} \cdot \bar{\bar{T}}_d \cdot \vec{n})\vec{n}]ds$.

Principal Directions of Strain (Notes.pdf, Eqs. 2.54–2.55)

There exist three mutually orthogonal spatial directions $(\vec{n}_1, \vec{n}_2, \vec{n}_3)$ along which shear strain vanishes identically (deformation consists purely of normal extension or compression). These are found by solving the eigenvalue problem:

$$\bar{\bar{T}}_d \cdot \vec{n} = \lambda \vec{n} \iff |\bar{\bar{T}}_d - \lambda \bar{\bar{I}}| = 0 \qquad [\text{Eqs. } 2.54, 2.55]$$

Since $\bar{\bar{T}}_d$ is real and symmetric, its eigenvalues $(\lambda_1, \lambda_2, \lambda_3)$ are real and its eigenvectors are mutually orthogonal.

Deformation of Fluid Elements: Volumetric Dilation

Considering the evolution of an infinitesimal cubic element of side length dl and initial volume $V_0 = dl^3$, the volume after time dt is given by the transformation determinant:

$$V_f = dl^3 \begin{vmatrix} 1 + \frac{\partial v_1}{\partial x_1} dt & \frac{\partial v_2}{\partial x_1} dt & \frac{\partial v_3}{\partial x_1} dt \\ \frac{\partial v_1}{\partial x_2} dt & 1 + \frac{\partial v_2}{\partial x_2} dt & \frac{\partial v_3}{\partial x_2} dt \\ \frac{\partial v_1}{\partial x_3} dt & \frac{\partial v_2}{\partial x_3} dt & 1 + \frac{\partial v_3}{\partial x_3} dt \end{vmatrix} \simeq dl^3 + dl^3 dt \left(\frac{\partial v_1}{\partial x_1} + \frac{\partial v_2}{\partial x_2} + \frac{\partial v_3}{\partial x_3} \right) \qquad [\text{Eq. } 2.62]$$

Physical Meaning of Velocity Divergence

Velocity divergence $\nabla \cdot \vec{v}$ represents the rate of volume change per unit volume (**volumetric expansion or dilation rate**) and equals the trace of $\nabla\vec{v}$ and $\bar{\bar{T}}_d$:

$$\mathbf{\frac{1}{V}\frac{dV}{dt} = \nabla \cdot \vec{v} = \mathrm{tr}(\bar{\bar{T}}_d) = \lambda_1 + \lambda_2 + \lambda_3}$$

For a perfect liquid (constant density), fluid particle volume cannot change:

$$\mathbf{\nabla \cdot \vec{v} = 0 \iff \frac{\partial v_1}{\partial x_1} + \frac{\partial v_2}{\partial x_2} + \frac{\partial v_3}{\partial x_3} = 0} \qquad [\text{Eq. 2.63}]$$

Consequently, for an incompressible fluid to deform while preserving volume, the extension rates $(\partial_1 v_1, \partial_2 v_2, \partial_3 v_3)$ **cannot all have the same sign**: any elongation in one direction must be balanced by compression in the others.

11. Comprehensive Formula Summary of Chapter 2

Kinematic Concept	Mathematical Expression	Equation in Notes.pdf
General metric differential	$d\vec{x} = (h_1 dx_1, h_2 dx_2, h_3 dx_3)$	p. 10
Gradient of a scalar	$\nabla\Phi = \left(\frac{1}{h_1}\frac{\partial\Phi}{\partial x_1}, \frac{1}{h_2}\frac{\partial\Phi}{\partial x_2}, \frac{1}{h_3}\frac{\partial\Phi}{\partial x_3}\right)$	Eq. 2.3
Divergence of a vector	$\nabla \cdot \vec{a} = \frac{1}{h_1 h_2 h_3}\sum_i \frac{\partial}{\partial x_i}\left(\frac{h_1 h_2 h_3}{h_i} a_i\right)$	Eq. 2.5
Curl of a vector	$\nabla \wedge \vec{a} = \frac{1}{h_1 h_2 h_3}\det[h_i\vec{e}_i, \partial/\partial x_i, h_i a_i]$	Eq. 2.6
Gauss divergence theorem	$\int_\Sigma (\vec{n} \circ \phi) d\sigma = \int_V (\nabla \circ \phi) dV$	Eq. 2.9
Fluid trajectory	$d\vec{x}/dt = \vec{v}(\vec{x}, t), \quad \vec{x}(t_0) = \vec{x}_0$	Eq. 2.12
Fluid surface	$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \vec{v} \cdot \nabla f = 0$	p. 14
Streamline	$\frac{h_1 dx_1}{v_1} = \frac{h_2 dx_2}{v_2} = \frac{h_3 dx_3}{v_3}$	Eq. 2.21
Material derivative	$\frac{D\phi}{Dt} = \frac{\partial\phi}{\partial t} + \vec{v} \cdot \nabla\phi$	Eq. 2.24
Intrinsic acceleration	$\vec{a} = \frac{\partial\vec{v}}{\partial t} + \nabla\left(\frac{\lvert\vec{v}\rvert^2}{2}\right) - \vec{v} \wedge (\nabla \wedge \vec{v})$	Eq. 2.26
Circulation and Vorticity	$\Gamma = \oint_L \vec{v} \cdot d\vec{l} = \int_\Sigma (\nabla \wedge \vec{v}) \cdot \vec{n} d\sigma, \quad \vec{\omega} = \nabla \wedge \vec{v}$	Eq. 2.32
Velocity potential	$\vec{\omega} = 0 \implies \vec{v} = \nabla\phi$	Eq. 2.34
Convective flux	$\Phi_{\text{conv}} = \int_\Sigma \phi\vec{v} \cdot \vec{n} d\sigma, \quad Q = \int_\Sigma \vec{v} \cdot \vec{n} d\sigma$	Eqs. 2.35, 2.36
Stream function	$v_x = \frac{\partial\psi}{\partial y},\ v_y = -\frac{\partial\psi}{\partial x}, \quad Q' = \psi_2 - \psi_1$	Eqs. 2.41, 2.45
Helmholtz decomposition	$\nabla\vec{v} = \bar{\bar{T}}_d + \bar{\bar{T}}_r, \quad d\vec{v}_r = \frac{1}{2}\vec{\omega} \wedge d\vec{x}$	Eqs. 2.47, 2.52
Volumetric dilation rate	$\frac{1}{V}\frac{dV}{dt} = \nabla \cdot \vec{v} = \mathrm{tr}(\bar{\bar{T}}_d)$	Eq. 2.62